

A Short Description of an Ankle-Actuated Seated VR Locomotion Interface

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ABSTRACT

We present an overview of the interface that allows users to perform foot-tapping movements in a seated posture to emulate walking sensation in virtual reality (VR). The interface comprises an impedance controlled robotic foot platform that moves while resisting the user's foot-tapping movements. The platform's angular movement is used to recognize the walking intention of the user and animate the VR avatar accordingly. The impedance characteristics of the platform are controlled to emulate different VR walking terrains.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices

1 INTRODUCTION

Virtual reality offers an opportunity to explore computer-generated environments for purposes such as gaming, architectural design and inspection or simulation training. Locomotion in virtual environments has been an interesting topic for researchers and developers. The main limitation of an immersive VR locomotion is that the user's workspace is much more confined than the one available in the virtual environment. Especially for home use, expanding the motion capture available area is generally not feasible due to space and financial issues. Using keyboards or joysticks are effective in first-person games played on a regular screen and some VR applications that do not emulate legged locomotion. However, they are known to cause simulator sickness due to limited sensorimotor feedback. Lack of body-centered interaction for VR locomotion causes spatial disorientation [11] and reduces immersion [12]. Point-and-teleport method [3] reduces the complexity in the required sensorimotor feedback and only requires basic hand controllers to operate. Currently, the point-and-teleport method is the most widely used method by general users but it completely abandons the natural walking sensation. To enable users to utilize their natural body movements, three categories of interfaces and methods are devised.

1. Repositioning systems like omnidirectional treadmills [4] which aim to physically move the user towards the middle of the workspace.

2. Redirection method [9] which tries to lead the user to the center of the room by scaling the mapping for translational and rotational motion in VR. Still requires a fairly large area or interruptions when the user gets close to the boundaries.

3. Walking-in-place (WIP) method uses walking-like gestures such as marching in place, arms swinging. They trade-off interface simplicity with the naturalness of the motion input.

We present a seated WIP type system, shown on Figure 1, that enables foot tapping movements to be used for moving inside VR

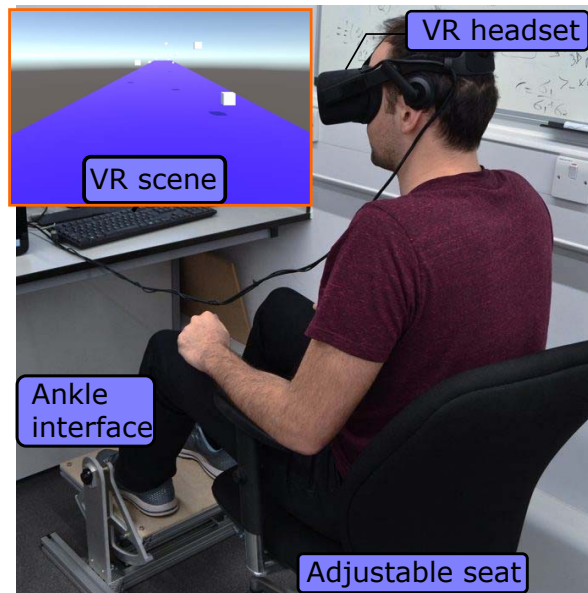


Figure 1: Ankle motion-based walking interface for seated VR

while retrieving terrain-related haptic feedback. We believe that terrain feedback can improve immersion and self localization. Haptic terrain feedback was implemented in standing interfaces [13, 16] but, to our knowledge, not in any seated interfaces. We also present a novel motion mapping algorithm designed for improving embodiment during VR locomotion WIP type interfaces.

2 RELATED WORK

This section presents previous walking-in-place interfaces and the advantages and challenges of using a seated walking-in-place interface for locomotion in VR.

Walking-in-Place type locomotion interfaces. Many studies have presented walking-in-place type systems which use variety of walking gestures such as marching in place [2, 6, 17], arm swinging [7], head movement [15] or foot sliding [14]. WIP interfaces generally focus on lowering the cost, space, and computational complexity requirements. Therefore full-body motion tracking is rarely desired. The data that is used to compute VR locomotion is either our tracking of the limb which performs the walking gesture or discrete events that are triggered at different stages of motion [10]. Interfaces that measure the user's movement continuously, recognize the walking gesture motion as a repetitive oscillatory signal. Algorithms like LLCM-WIP [6] and GUD-WIP [17] compute avatar

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forward velocity from this oscillatory signal in real-time. While LLCM-WIP uses the absolute value of the derivative of the gesture input to compute the avatar walking speed, GUD-WIP computes the frequency of the oscillatory signal and computes the avatar walking speed accordingly. Our latest algorithm, presented in Section 3.3, is similar to the latter one but we compute the avatar's walking cycle phase to be able to utilize an anthropomorphic avatar for improving the sense of body ownership.

Seated interaction. Seated interfaces generally provide higher comfort and upper body stability for carrying out manual tasks. We categorize lower-body driven seated interfaces as (1) wearable interfaces like Cybershoes [1] which allows the user to rotate around on a swivel chair to steer the avatar and (2) stationary interfaces like Pointman [14] and ours which require the user's body to be always facing the direction of the interface and use an extra method to perform the steering movement. Other seated interfaces that require the user to be facing a certain direction, like towards a webcam in Terziman et. al. [15], have the same limitations as the latter category. To tackle this challenge Pointman [14] utilizes can be using a joystick or keyboard to steer or direct the forward movement motion to steering. Terziman et. al. [15] uses upper body inclination to simulate whole body inclination while steering. Templeman, in his patent [5], suggested using sideways foot sweeping. We have decided to utilize method presented by Terziman et. al. since it utilizes the headset IMU which is still a body-based interaction and does not require us to add an extra degree of freedom to feet movement detection.

3 WALKING INTERFACE

In this section, we explain the structural components of the system, control architecture, working principle, motion mapping algorithm and haptic terrain feedback.

3.1 System Architecture

The hardware part of the interface consists of a single capstan driven 1-DOF foot platform, a backdrivable brushed DC motor, control electronics, a height adjustable chair and a commercial VR headset (Oculus Rift CV1) as shown in Figure 1. The platform is rigidly connected to a chair for a more stable interaction. The users are expected to sit on the chair, place their feet on the platform and perform ankle dorsiflexion and plantarflexion movements while keeping their heels stationary. The seat location is adjusted to fit the body size of the user by sliding the chair along the connection to the platform and using the height adjustment of the chair. The swiveling feature of the chair is not utilized.

A geared back-driveable DC motor actuates the platform through capstan transmission. Platform angle and angular velocity, which are essential to detecting the feet movement, are measured using the optical shaft encoder connected to the DC motor. A proportional-derivative(PD) controller along with gravity compensation is used to control the platform angle. The platform is allowed to rotate $\pm 20^\circ$ around its neutral (horizontal) position. Dorsiflexion of the foot causes platform movement in the positive direction, whereas plantar flexion causes movement in the negative direction. Our earlier publication [8], focused on design and control of the platform, can provide more insight on its mechanical properties.

The platform angle data is used to calculate the forward movement of the avatar. The avatar movement data was communicated to a Unity 3D application where the user's avatar is animated accordingly. A desktop computer is used to communicate between the data acquisition device and the virtual reality scene. The VR headset provides the user visual feedback and provides the Unity application user's head orientation, which is used for avatar steering.

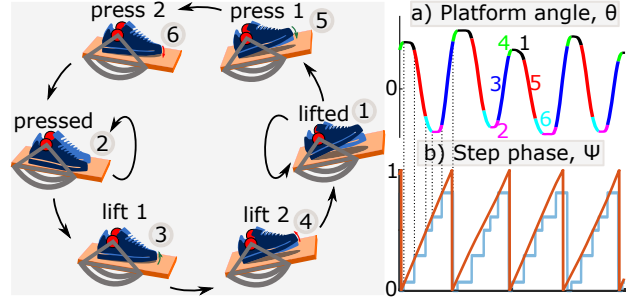


Figure 2: **Left:** stages of movement for a single foot with region numbers written in gray circles. **Right:** a sample platform angle to step phase mapping using the presented algorithm.

3.2 Proposed feet movement

Users are required to make stepping gestures comprising consequent left/right ankle plantar/dorsiflexion movements to achieve walking in the VR scene. Such repetitive stepping gestures correspond to physical ankle movements during natural walking. The periodic left/right ankle flexion gestures can be efficiently realized with a single actuated robotic platform. For an efficient locomotion while continuously stepping, a smooth transition between right and left foot movement is required similar to how natural walking requires a smooth load transfer between legs. If the left and right feet perform opposite extension movements at the same time they negate each other due to using a single foot platform rather than double. Hence, using a single platform enforces a rhythmic feet movement.

3.3 Avatar forward movement

In this section, we present our most recent version of the walking algorithm which is designed to simulate the full body walking animation of an avatar. The previous algorithm we worked on, along with many of the presented algorithms in the literature, only compute the avatar forward velocity. The presented algorithm, improves on top of previous algorithms to capture the gait pattern better. The motion input should be measured at high sampling rate and low noise levels for attaining better derivative signals. Our interface was designed to provide such an input but other interfaces may experience a decay in performance with the same algorithm. Our new algorithm aims to compute the exact phase of the stepping movement so that any VR avatar with a suitable walking animation can be utilized. By matching the users gesture movement to the walking cycle of the anthropomorphic avatar, we aim to improve the sense of embodiment the user feels. Our algorithm divides the platform angle data, θ and its derivatives to divide the stepping motion into six regions as listed below and depicted on Figure 2.

- stopping regions: $|\dot{\theta}| < \omega_0$
 1. when foot is lifted: $\dot{\theta} > 0$ in the previous region.
 2. when foot is pressed: $\dot{\theta} < 0$ in the previous region.
- movement regions: $|\dot{\theta}| > k\omega_0$, where $n > 1$
 3. accelerating upwards: $\dot{\theta} > 0, \ddot{\theta} > 0$
 4. decelerating upwards: $\dot{\theta} > 0, \ddot{\theta} < 0$
 5. accelerating downwards: $\dot{\theta} < 0, \ddot{\theta} < 0$
 6. decelerating downwards: $\dot{\theta} < 0, \ddot{\theta} > 0$

The algorithm detects every time a transition from one region to another occurs. The aim of the algorithm is to estimate the step phase between the transition events, presented by dotted lines in

Figure 2, and correct for the error as the transition events occur. A step phase variable is introduced to compute the gait phase for the avatar's legs' motion based on the identified phase transitions.

3.4 Steering

Walking in 3D requires us to devise a steering method for the avatar. We use sideways head tilting to command avatar steering in VR. We selected this movement since head pitch movement is not as much utilized as roll and yaw movements. The head-mounted display's IMU sensors capture the necessary data to compute steering during locomotion. The pitch angle of a user's head orientation from straight orientation is proportionally mapped to the longitudinal angular velocity of the avatar. We also used a deadzone function to prevent pitch angles below a threshold from causing steering; and a saturation function to prevent pitch angles above a threshold from causing faster steering to avoid cybersickness.

3.5 Haptic terrain feedback

The purpose behind designing the interface as an actively controlled platform, rather than one that has fixed compliance, is to be able to alternate the impedance depending on the VR terrain. We believe that terrain information is crucial in helping the users understand their self-motion and supporting their visual sense of motion. We briefly describe three terrain types that can be rendered by the interface.

Stiff rigid terrain. This terrain model implements the model of a virtual wall with fixed stiffness and damping. The position of the virtual wall which actually represents the virtual terrain is set to horizontal orientation of the ankle platform.

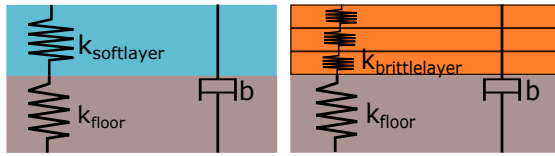


Figure 3: **Left:** a depiction of single layer possessing softer stiffness placed on the floor layer. **Right:** number of high stiffness but low strength layers placed on the floor layer.

Soft layered terrain. This soft layer was modeled as an addition on top of the stiff terrain. As depicted in Figure 3, the softer layer is modeled as a softer spring and the same damping element. The springs are not serially connected and the stiff spring comes into effect once the soft layer is surpassed. The force profile is kept continuous by modeling the stiff terrain in a pre-tensioned state.

Brittle layered/crunchy terrain. We model this terrain as a serial connection of several layers of spring and damper systems added on top of the stiff terrain as shown in Figure 3. The springs in the brittle layers were set up to be stiffer than the stiff/ground layer. Each brittle layer behaves like brittle materials which would undergo linear elastic deformation until they break after a certain compressing strain, aligned with Mohr-Coulomb yield criterion. The materials were designed to crack at slightly different compressive stress and strain values since we want them to crack one after the other. At the instance when a layer cracks, it shrinks to around half of its original size, and its stiffness is doubled. A temporary relaxation after breaking a layer occurs and it is followed by a stiffer resistance. As the layers collapse the cumulative rest length of the springs drop. To tackle this, a nonlinear spring with the original rest length is applied after the foot starts lifting to lead the platform above the horizontal level so that the brittle layers can be reloaded for the next step.

4 CONCLUSION

A seated VR interface with an algorithm based on feet gestures for locomotion control of anthropomorphic VR avatars is presented.

The proposed approach associates the user's foot tapping and upper body inclination movements to readily available walking animations to enhance the sense of immersion and embodiment of the VR locomotion task.

Our preliminary trials, which required users to walk towards and stop at target in VR, showed all users were comfortable with walking forward to reach the target but when steering was involved, some of the users suffered cybersickness. A further comparative study on VR steering techniques on a seated interface, mentioned in Section 3.4, can provide valuable insight for avoiding cybersickness.

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REFERENCES

- [1] *CyberShoes VR locomotion interface*.
- [2] L. Bouguila, B. Hirsbrunner, M. Sato, and M. Iwashita. Virtual locomotion interface with ground surface simulation. In *ICAT*, 2003.
- [3] E. Bozgeyikli, A. Raij, S. Katkooi, and R. Dubey. Point & teleport locomotion technique for virtual reality. In *Proceedings of the 2016 Annual Symposium on Computer-Human Interaction in Play*, CHI PLAY '16, p. 205–216. Association for Computing Machinery, New York, NY, USA, 2016. doi: 10.1145/2967934.2968105
- [4] R. P. Darken, W. R. Cockayne, and D. Carmein. The omni-directional treadmill: A locomotion device for virtual worlds. In *Proceedings of the 10th Annual ACM Symposium on User Interface Software and Technology*, UIST '97, pp. 213–221. ACM, New York, NY, USA, 1997. doi: 10.1145/263407.263550
- [5] T. J. Edison. Simulated locomotion method and apparatus, U.S. Patent 7,542,040 B2, June 2009.
- [6] J. Feasel, M. C. Whitton, and J. D. Wendt. LLCM-WIP: Low-latency, continuous-motion walking-in-place. In *2008 IEEE Symposium on 3D User Interfaces*, pp. 97–104, March 2008.
- [7] N. C. Nilsson, S. Serafin, and R. Nordahl. The perceived naturalness of virtual locomotion methods devoid of explicit leg movements. In *Proceedings of Motion on Games*, MIG '13, pp. 133:155–133:164. ACM, New York, NY, USA, 2013. doi: 10.1145/2522628.2522655
- [8] A. Otaran and I. Farkhatdinov. Modeling and control of ankle actuation platform for human-robot interaction. In *Towards Autonomous Robotic Systems*, pp. 338–348. Springer International Publishing, 2019.
- [9] S. Razzaque, Z. Kohn, and M. C. Whitton. Redirected Walking. In *Eurographics 2001 - Short Presentations*. Eurographics Association, 2001.
- [10] P. Richard, L. Bouguilla, M. Courant, and B. Hirsbrunner. Enactive navigation in virtual environments: Evaluation of the walking pad. 2007.
- [11] R. A. Ruddle and S. Lessels. For efficient navigational search, humans require full physical movement, but not a rich visual scene. *Psychological Science*, 17(6):460–465, 2006. PMID: 16771793.
- [12] M. Slater and M. Usoh. Body centred interaction in immersive virtual environments. 1994.
- [13] H. Son, I. Hwang, T. Yang, S. Choi, S. Kim, and J. R. Kim. Realwalk: Haptic shoes using actuated mr fluid for walking in vr. In *2019 IEEE World Haptics Conference (WHC)*, pp. 241–246, 2019.
- [14] J. N. Templeman, L. E. Sibert, R. C. Page, and P. S. Denbrook. Pointman - a new control for simulating tactical infantry movements. In *2007 IEEE Virtual Reality Conference*, pp. 285–286, March 2007.
- [15] L. Terziman, M. Marchal, M. Emily, F. Multon, B. Arnaldi, and A. Lécuyer. Shake-your-head: revisiting walking-in-place for desktop virtual reality. In *VRST*, 2010.
- [16] Y. Visell, J. R. Cooperstock, B. L. Giordano, K. Fratinovic, A. Law, S. McAdams, K. Jathal, and F. Fontana. A vibrotactile device for display of virtual ground materials in walking. In M. Ferre, ed., *Haptics: Perception, Devices and Scenarios*, pp. 420–426. Springer Berlin Heidelberg, Berlin, Heidelberg, 2008.
- [17] J. D. Wendt, M. C. Whitton, and F. P. J. Brooks. GUD-WIP: Gait-understanding-driven walking-in-place. *Proceedings. IEEE Virtual Reality Conference*, 2010:51–58, Mar 2010. 25621313[pmid].